CS 3110/5110: Data Privacy

In-Class Exercise, Week of 9/29/2025

```
# Load the data and libraries
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

def laplace_mech(v, sensitivity, epsilon):
    return v + np.random.laplace(loc=0, scale=sensitivity / epsilon)

def laplace_mech_vec(vec, sensitivity, epsilon):
    return [v + np.random.laplace(loc=0, scale=sensitivity / epsilon)
for v in vec]

def gaussian_mech(v, sensitivity, epsilon, delta):
    return v + np.random.normal(loc=0, scale=sensitivity *
np.sqrt(2*np.log(1.25/delta)) / epsilon)

def gaussian_mech_vec(vec, sensitivity, epsilon, delta):
    return [v + np.random.normal(loc=0, scale=sensitivity *
np.sqrt(2*np.log(1.25/delta)) / epsilon) for v in vec]

def pct_error(orig, priv):
    return np.abs(orig - priv)/orig * 100.0

adult =
pd.read_csv('https://github.com/jnear/cs3110-data-privacy/raw/main/
homework/adult_with_pii.csv')
retail =
pd.read_csv('https://github.com/jnear/cs3110-data-privacy/raw/main/
homework/retail.csv')
```

Question 1

Implement the Gaussian mechanism for (ϵ, δ)-differential privacy.

```
# YOUR CODE HERE
def gaussian_mech(v, senditivity, epsilon, delta):
    sigma_squared = 2*senditivity**2 * np.log(1.25 / delta) /
(epsilon**2)
    sigma = np.sqrt(sigma_squared)
    return v + np.random.normal(loc=0, scale=sigma)
```

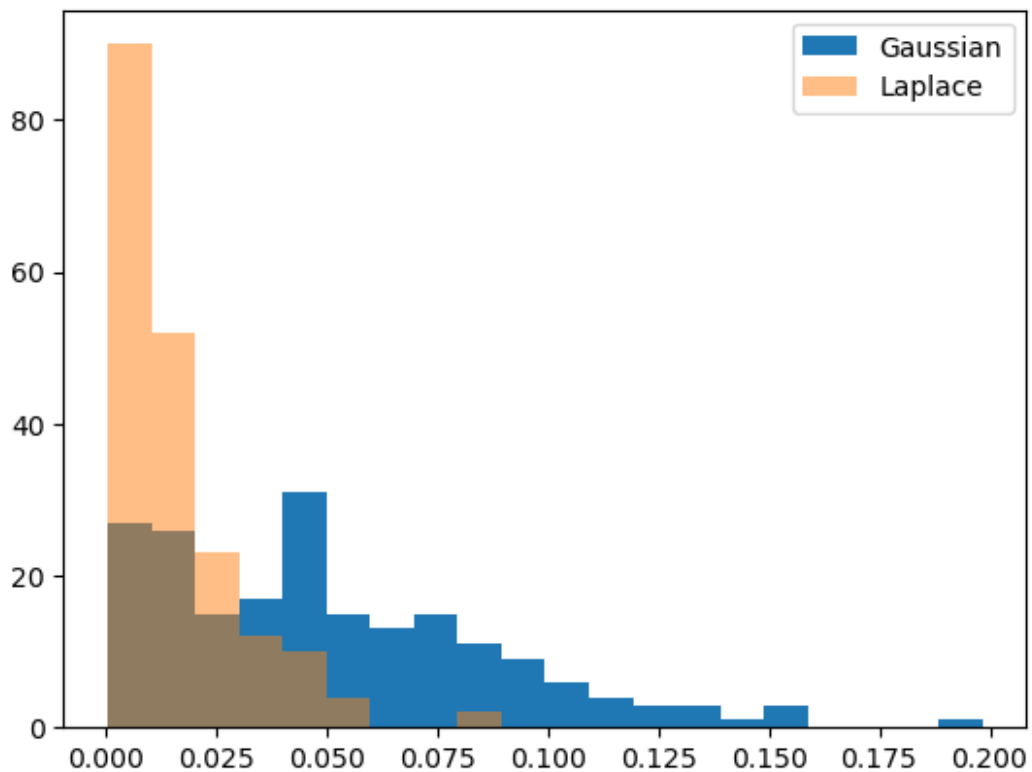
```
# gaussian_mech(len(adult), 1, epsilon=1.0, delta=1e-5)
# delta typically very small, like 10^-5

# TEST CASE

results = [gaussian_mech(len(adult[adult['Age'] > 50]), 1, 1.0, 10e-5)
for _ in range(100)]
errors = [pct_error(len(adult[adult['Age'] > 50]), r) for r in
results]
print('mean error:', np.mean(errors))

assert np.mean(errors) > 0
assert np.mean(errors) < 2
```

```
mean error: 0.060046527664626294
```

Question 2

How do the Laplace and Gaussian mechanisms compare in terms of relative error on the query "how many individuals are over 50 years old" with $\epsilon = 1$ and $\delta = 10^{-5}$?

```
true_answer = len(adult[adult['Age'] > 50])

laplace_answers = [laplace_mech(true_answer, 1, 1) for _ in
range(200)]
gaussian_answers = [gaussian_mech(true_answer, 1, 1, 10e-5) for _ in
range(200)]

laplace_error = [pct_error(true_answer, a) for a in laplace_answers]
gaussian_error = [pct_error(true_answer, a) for a in gaussian_answers]

_, bins, _ = plt.hist(gaussian_error, bins=20, label='Gaussian')
plt.hist(laplace_error, bins=bins, label='Laplace', alpha=0.5)
plt.legend();
```



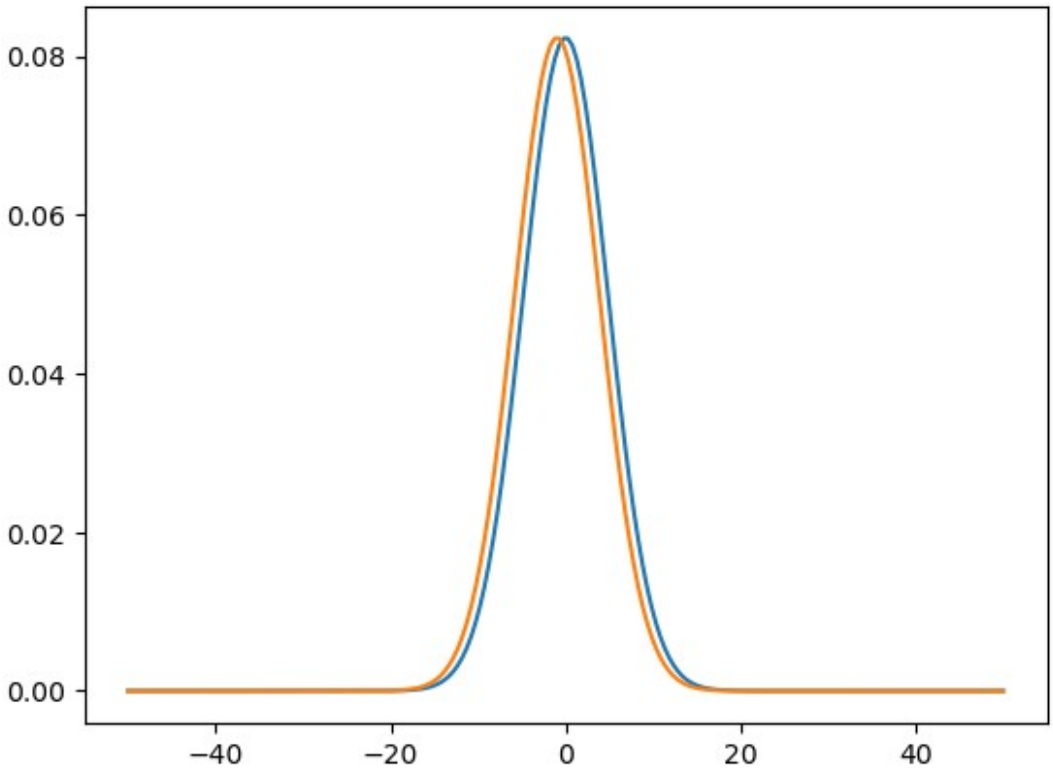
The Laplace mechanism has better relative error consistently.

Not a Question - Just for reference

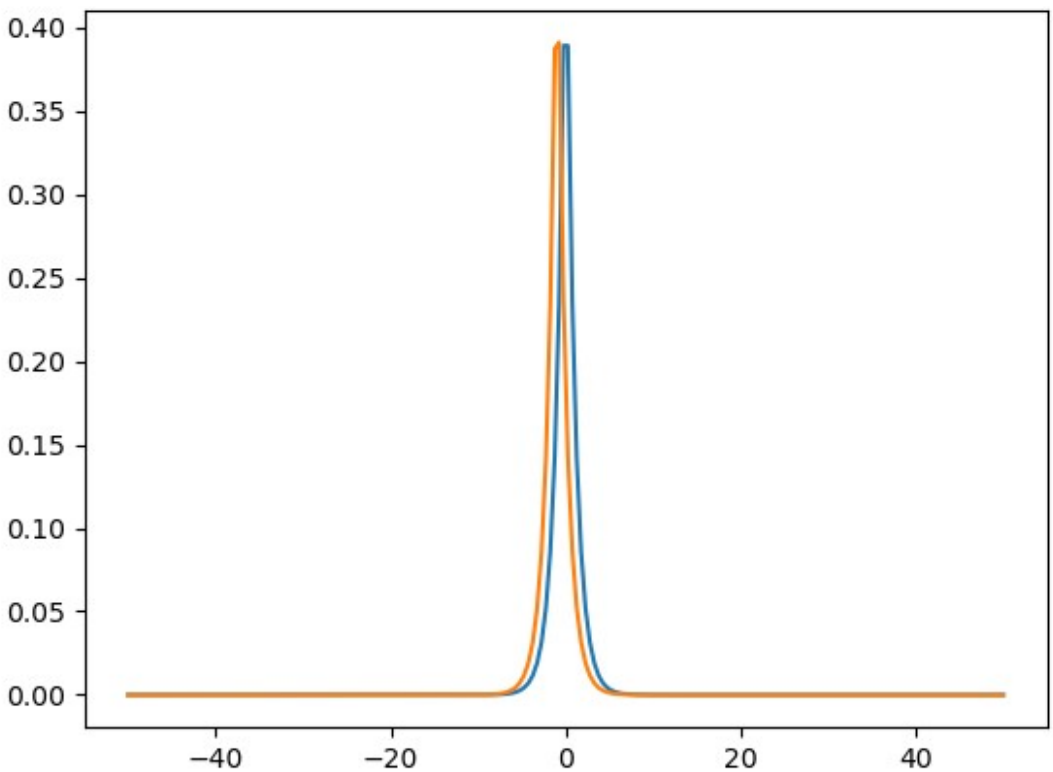
Reference

```
epsilon = 1
sensitivity = 1
delta = 1e-5
sigma_squared = 2 * sensitivity**2 * np.log(1.25 / delta) /
(epsilon**2)
sigma = np.sqrt(sigma_squared)

def gauss_pdf(x):
    return 1/(sigma*np.sqrt(2*np.pi)) * np.exp(-(1/2)*(x/sigma)**2)

xs = np.linspace(-50, 50, 200)
ys1 = [gauss_pdf(x) for x in xs]
ys2 = [gauss_pdf(x+1) for x in xs]

plt.plot(xs,ys1)
plt.plot(xs,ys2)

# ratio < e^epsilon should hold
print('e^epsilon =', np.exp(epsilon))
```

```
ratios = [(x, y1 / y2) for x, y1, y2 in zip(xs, ys1, ys2)]
#ratios

e^epsilon = 2.718281828459045
```




```
def laplace_pdf(x):
    return (1/2)*epsilon * np.exp(-np.abs(x)*epsilon)

xs = np.linspace(-50, 50, 200)
ys1 = [laplace_pdf(x) for x in xs]
ys2 = [laplace_pdf(x+1) for x in xs]

plt.plot(xs,ys1)
plt.plot(xs,ys2)

# ratio < e^epsilon should hold
print('e^epsilon =', np.exp(epsilon))
ratios = [(x, y1 / y2) for x, y1, y2 in zip(xs, ys1, ys2)]
#ratios

e^epsilon = 2.718281828459045
```



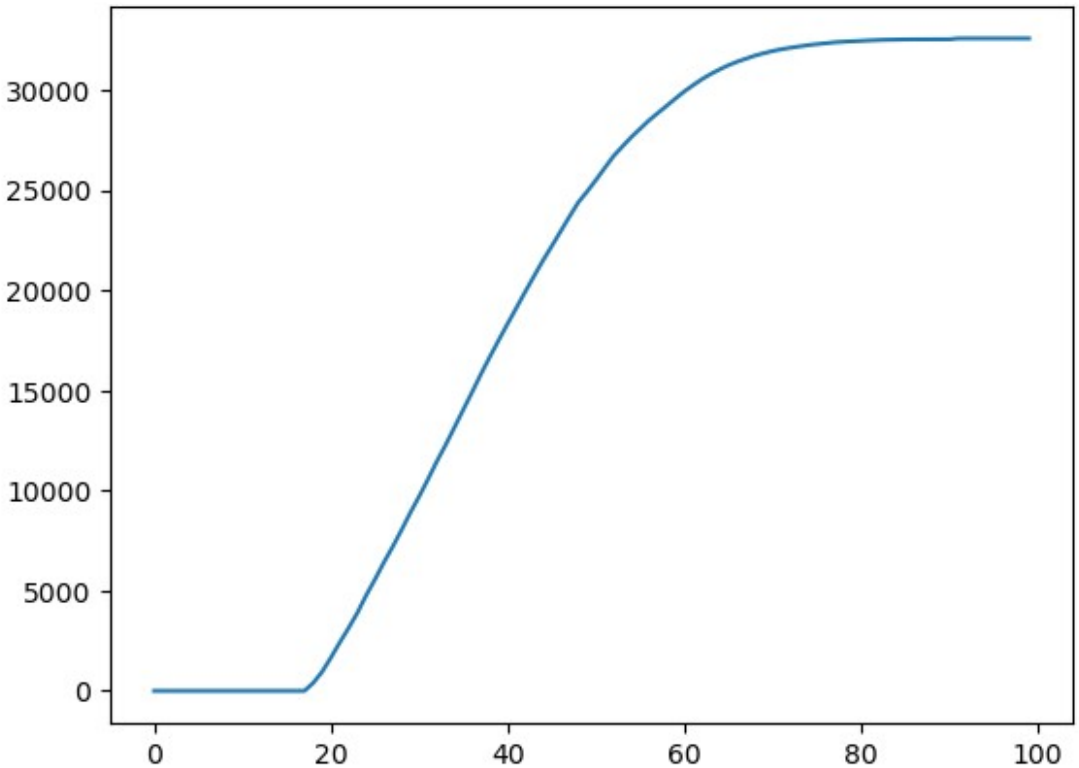
Question 3

The function `age_cdf` calculates the cumulative distribution function (CDF) of the ages in the adult dataset, expressed as a vector of counts.

What is the **L1 global sensitivity** of `age_cdf`?

```
def age_cdf():
    a = adult['Age']
    return [len(a[a < i]) for i in range(100)]

plt.plot(age_cdf());
print('Length of CDF vector:', len(age_cdf()))
```

```
Length of CDF vector: 100
```



This function returns a vector of 100 values.

I can't use parallel composition when adding noise to these values, because the counts in the vector do not come from disjoint subsets of the data (one person might appear in multiple counts).

The L1 global sensitivity of age_cdf is the sum of the element-wise sensitivities of the vector.

Each element of the vector has sensitivity 1 because it's a counting query.

Thus, the L1 global sensitivity is 1 + 1 + 1 + … (100 times) = 100.

Question 4

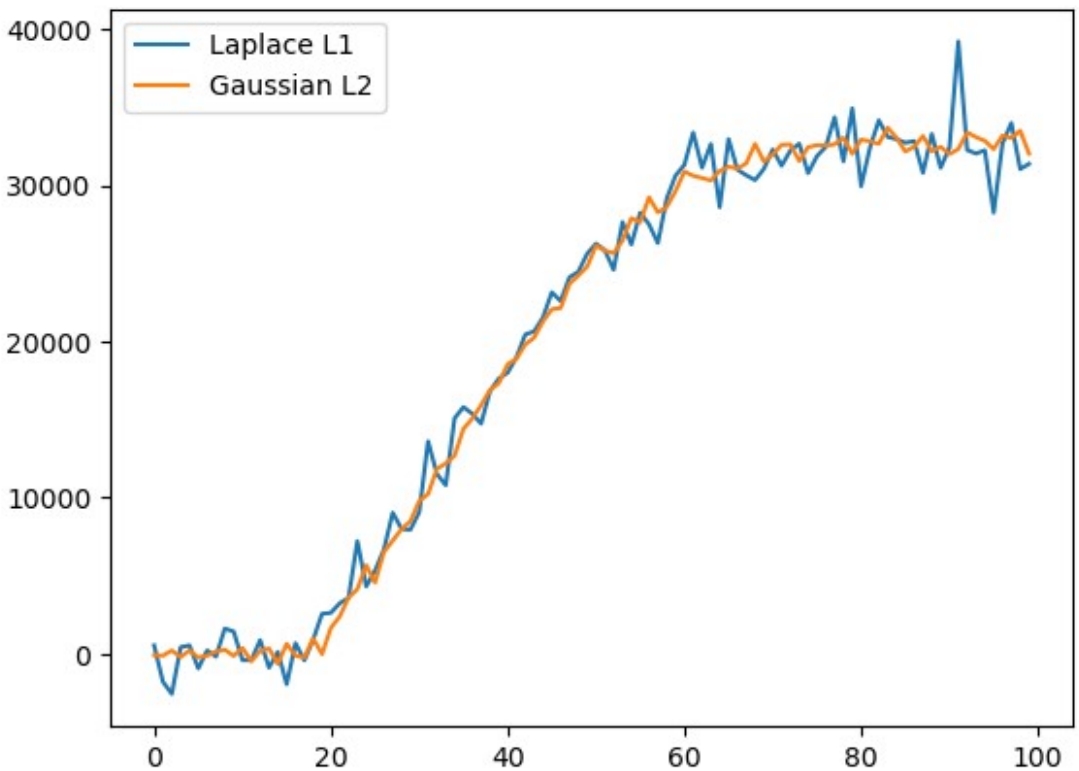
What is the **L2 global sensitivity** of `age_cdf`?

- The L2 global sensitivity is the square root of the sum of the squares of the element-wise sensitivities.
- This is sqrt(1^2 + 1^2 + … + 1^2) = sqrt(100) = 10.

Question 5

Use the vector-valued Laplace and Gaussian mechanisms to construct noisy versions of the CDF of ages. **Be sure to use the right sensitivity!**

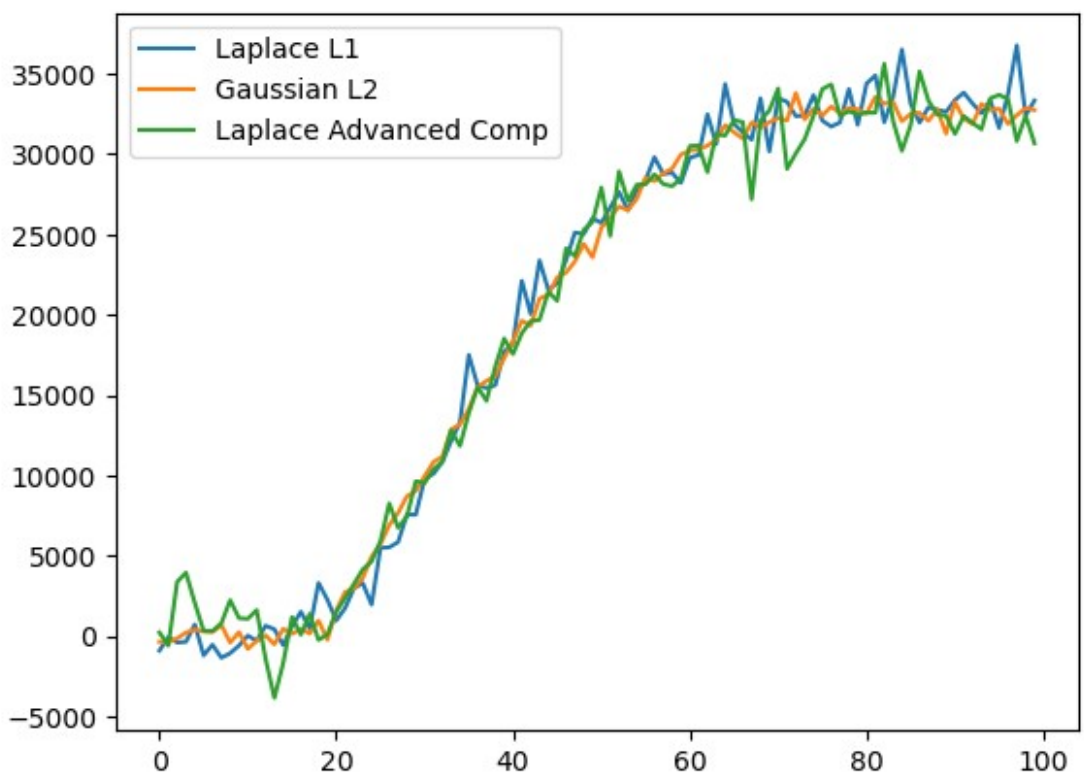
```
def laplace_cdf(epsilon):
    l1_sens = 100
    # laplace mech (vector) adds noise to a vector with the given L1
sensitivity
    # it satisfies a total privacy cost of epsilon DP without any
composition
    return laplace_mech_vec(age_cdf(), sensitivity=l1_sens,
epsilon=epsilon)

def gaussian_cdf(epsilon, delta):
    l2_sens = 10
    return gaussian_mech_vec(age_cdf(), sensitivity=l2_sens,
epsilon=epsilon, delta=delta)

plt.plot(laplace_cdf(0.1), label='Laplace L1');
plt.plot(gaussian_cdf(0.1, 1e-5), label='Gaussian L2');
plt.legend();
```



Question 6

Use the *regular* (not vector-valued) Laplace mechanism, **with advanced composition**, to construct a noisy CDF of ages.

```
def laplace_cdf_advanced_composition(epsilon, delta):
    # wrong answer: uses sequential composition
    a = adult["Age"]
    epsilon_1 = epsilon/100
    noisy_cdf = []
    for i in range(100):
        # sensitivity = 1 because this is a counting query returning a
single number (not a vector)
        noisy_count = laplace_mech(len(a[a<i]),
sensitivity=1,epsilon=epsilon_1)
        noisy_cdf.append(noisy_count)
    return noisy_cdf
# total privacy cost: 100 iterations * epsilon_1 = 100 * (epsilon/100)
= epsilon
# by sequential composition

plt.plot(laplace_cdf(0.1), label='Laplace L1')
plt.plot(gaussian_cdf(0.1, 1e-5), label='Gaussian L2')
plt.plot(laplace_cdf_advanced_composition(0.1, 1e-5), label='Laplace
Advanced Comp')
plt.legend();
```



Question 7

Consider the retail dataset and the following counting query on it. What is the query's sensitivity?

```
def retail_query():
    return len(retail[retail['CustomerID'] == 12748])

retail_query()
```

YOUR ANSWER HERE

Question 8

Use contribution bounding to adjust the unit of privacy to be person-level privacy, and implement a differentially private function to return the orders originating from the United Kingdom.

```
def bound_contributions(bound):
    # YOUR CODE HERE
    raise NotImplementedError()

def dp_retail_query(bound, epsilon):
    # YOUR CODE HERE
    raise NotImplementedError()

dp_retail_query(100, 1.0)

# TEST CASE

results = [dp_retail_query(100, 1.0) for _ in range(100)]
errors = [pct_error(len(retail[retail['Country'] == 'United
Kingdom']), r) for r in results]
print('mean error:', np.mean(errors))

assert np.mean(errors) > 20
assert np.mean(errors) < 50
```